CHAPTER 12

Monte Carlo Simulation & Quantitative Risk Analysis

Vectorized Stochastic Ensembles, The Central Limit Theorem, Goal Attainment Probabilities, and Value at Risk

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12.1 The Paradigm Shift: From Point Estimates to Probability Densities

Traditional retail calculators suffer from **deterministic expectation bias**: they present investors with a single, static future corpus assuming zero market volatility. In reality, market outcomes form an entire **probability distribution**. **Monte Carlo simulation** bridges this gap by generating thousands of independent stochastic price trajectories, revealing downside tail risks, median paths, and empirical goal attainment odds.

Definition 12.1 (Monte Carlo Estimator & Law of Large Numbers)

Let M be the number of independent simulated market paths, and let $S_{T,\,m}$ denote the terminal asset price for path $m \in \{1, 2, ..., M\}$.
By the **Strong Law of Large Numbers (SLLN)**, the empirical sample mean converges almost surely to the true expectation:

$$\lim_{M \to \infty} (1 / M) \sum_{m=1}^{M} g(S_{T,\,m}) = \blacksquare[g(S_T)] \quad \text{(a.s.)}$$

*The **Target Goal Attainment Probability** for financial target Γ is $\Phi\blacksquare(\Gamma) = (1/M) \sum_{m=1}^{M} \blacksquare_{\{S_{T,\,m} \geq \Gamma\}}$.*

12.2 Convergence Rate & Asymptotic Standard Error

The Central Limit Theorem (CLT) Error Bound:
Let $\sigma\blacksquare_g$ denote the sample standard deviation of terminal simulated outcomes. By the Central Limit Theorem, the statistical error of the Monte Carlo estimator decays strictly at rate:

- **Standard Error:** SE = $\sigma\blacksquare_g / \sqrt{M} = O(M^{-1/2})$.
- **Dimension Independence:** Unlike deterministic numerical quadrature (whose error explodes exponentially in higher dimensions as $O(N^{-1/d})$), the Monte Carlo convergence rate $O(M^{-1/2})$ is **strictly independent of dimension**. This makes simulation the preeminent tool for multi-asset portfolios and stochastic paths.

Theorem 12.1 (Quantitative Risk Metrics: Percentiles & Value at Risk)

For terminal wealth distribution vector $\mathbf{W}_T = \{W_{T,\,1}, ..., W_{T,\,M}\}$:

- **Median Expected Wealth:** P_{50} = Percentile($\mathbf{W}_T$, 50)
- **Downside Bear Scenario (5th Percentile):** P_5 = Percentile($\mathbf{W}_T$, 5)
- **Empirical Value at Risk (VaR at 95% Confidence):**

$\mathbf{VaR_{95\%} = W_0 - P_5}$ **(or relative: $\boldsymbol{P_{50} - P_5}$)**

Decision Support Value: VaR quantifies the maximum capital drawdown expected under 95% of all market states, allowing investors to stress-test financial plans before committing capital.

12.3 Step-by-Step Worked Examples

Below are two formal case studies solved by hand: analytical goal attainment probability estimation, and empirical 10,000-path statistical error bounds with Value at Risk.

Example 12.1: Analytical Target Goal Attainment Probability

Problem Statement: An investor commits S_0 **= INR 1,000,000** into a portfolio with expected drift μ **= 12.00%** and volatility σ **= 20.00%** for t **= 10 years**. The investor sets an ambitious milestone target of Γ **= INR 5,000,000** (5x capital replication). Calculate the exact analytical probability $P(S_{10} \geq \Gamma)$ and contrast with deterministic projections.

Step-by-Step Analytical Solution:

Step 1 (Transform to Lognormal Standard Normal Variable):

From Chapter 11, $\ln(S_t) \sim \blacksquare[\ \ln(S_0) + (\mu - \sigma^2/2)t,\quad \sigma^2 t\ ]$.

- Expected log mean: $\mu_{log} = \ln(1{,}000{,}000) + [0.12 - (0.04/2)] \times 10 = 13.815511 + (0.10 \times 10) =$ **14.815511**.
- Log variance: $\sigma_{log}^2 = (0.20)^2 \times 10 = 0.4000 \Rightarrow$ Standard Deviation: $\sigma_{log} = \sqrt{0.40} \approx$ **0.6324555**.

Step 2 (Evaluate Target Logarithm & Z-Score):

- Target log: $\ln(\Gamma) = \ln(5{,}000{,}000) \approx$ **15.424949**.
- Standard normal Z-statistic:

$Z = [\ \ln(\Gamma) - \mu_{log}\ ] / \sigma_{log} = [\ 15.424949 - 14.815511\ ] / 0.6324555 = 0.609438 / 0.6324555 \approx$ **+0.963606**.

Step 3 (Probability Computation via Standard Normal CDF Φ):

- $P(S_{10} \geq 5{,}000{,}000) = 1 - \Phi(0.9636)$.
- From standard normal tables: $\Phi(0.9636) \approx 0.83238$.
- **Target Attainment Probability:** 1 – 0.83238 = 0.16762 &implies; **16.76% success odds**.

Step 4 (Deterministic Fallacy vs. Stochastic Reality):

- Deterministic Calculator Projection: $1{,}000{,}000 \times (1.12)^{10}$ = INR 3,105,848 (or continuous INR 3,320,117).
- *Crucial Insight:* A static calculator provides no probability metric. An investor targeting INR 5 Million might mistakenly believe their target is reasonably achievable, when the empirical mathematical reality reveals an **83.24% failure risk!**

Example 12.2: 10,000-Path Simulation Statistics & Value at Risk (VaR)

Problem Statement: A 10,000-path Monte Carlo simulation is executed on an equity investment of S_0 **= INR 1,000,000** over a t **= 1-year horizon** with drift μ **= 14.0%** and volatility σ **= 25.0%**. The sample statistics return sample mean M■ = INR 1,148,500, sample standard deviation s = INR 292,000, and 5th percentile P_5 = INR 724,100. Calculate the 95% Confidence Interval for the mean and the 1-year Value at Risk ($VaR_{95\%}$).

Step-by-Step Analytical Solution:

Step 1 (Compute Central Limit Theorem Standard Error):

- Number of simulation paths: $M = 10{,}000 \Rightarrow \sqrt{M} = \sqrt{10{,}000} = 100$.
- Standard Error: SE = $s / \sqrt{M}$ = 292,000 / 100 = **INR 2,920.00**.

Step 2 (Construct 95% Confidence Interval for Expected Mean):

- Margin of error: 1.96 × SE = 1.96 × 2,920.00 = **INR 5,723.20**.
- 95% Confidence Interval: [1,148,500 – 5,723.20, 1,148,500 + 5,723.20] = **[INR 1,142,776.80, INR 1,154,223.20]**.
- Theoretical analytical mean: 1,000,000 × $e^{0.14}$ = **INR 1,150,273.80** (lies perfectly within CI!).

Step 3 (Calculate Absolute Value at Risk — $VaR_{95\%}$):

- 5th percentile terminal value: P_5 = INR 724,100.00.
- Absolute Dollar VaR: $VaR_{95\%} = S_0 - P_5$ = 1,000,000 – 724,100 = **INR 275,900.00**.
- Percentage VaR: 275,900 / 1,000,000 = **27.59% maximum expected drawdown**.

Step 4 (Risk Interpretation for Decision Support):

- *Takeaway:* In 95% of all market years, the investor's maximum capital loss will not exceed INR 275,900. However, in 5% of severe bear-market years, capital losses will be even more severe. This empowers the investor to align risk tolerance before purchasing the asset.

12.4 Algorithmic Implementation in Python

Below is the complete vectorized Monte Carlo simulation engine from models/monte_carlo.py, generating full path ensembles, empirical percentiles, and goal attainment probability estimators:

Listing: models/monte_carlo.py — Vectorized Simulation & Risk Engine

```
import numpy as np
from typing import Dict

def simulate_monte_carlo_portfolio(
    s0: float, mu: float, sigma: float, time_years: float,
    n_simulations: int = 10000, target_goal: float = None, seed: int = None
) -> Dict:
    """
    Executes vectorized Monte Carlo simulation under Geometric Brownian Motion.
    Returns mean, median, percentiles (5th, 95th), VaR_95, and target probability.
    """
    if s0 <= 0.0 or sigma < 0.0 or time_years <= 0.0 or n_simulations <= 0:
        raise ValueError("Simulation parameters must be strictly positive.")
    if seed is not None:
        np.random.seed(seed)
    drift = (mu - 0.5 * (sigma ** 2)) * time_years
    diffusion = sigma * np.sqrt(time_years)
    z = np.random.normal(0.0, 1.0, size=n_simulations)
    terminal_values = s0 * np.exp(drift + diffusion * z)
    p5, p50, p95 = np.percentile(terminal_values, [5.0, 50.0, 95.0])
    target_prob = float(np.mean(terminal_values >= target_goal)) if target_goal else None
    return {
        'mean': float(np.mean(terminal_values)),
        'median': float(p50),
        'p5': float(p5),
        'p95': float(p95),
        'var_95_abs': float(max(0.0, s0 - p5)),
        'target_probability': target_prob,
        'standard_error': float(np.std(terminal_values) / np.sqrt(n_simulations))
    }
```

12.5 Mathematical Properties Summary

Mathematical Property	Specification & Formula	Analytical Significance
LLN Convergence	lim M■ = ■[S_T]	Almost sure convergence of sample ensemble to true expectation
CLT Error Rate	SE = σ / √M = $O(M^{-1/2})$	Dimension-independent convergence rate for stochastic integrals
Goal Probability	P■(Γ) = (1/M)∑■$_{\{S_T \geq \Gamma\}}$	Unbiased empirical estimator of milestone achievement likelihood
Value at Risk	$\text{VaR}_{95\%} = S_0 - P_5$	Quantifies maximum capital drawdown at 95% confidence
Vectorization Speed	NumPy array operations	High-throughput SIMD parallelization over 10,000+ paths